

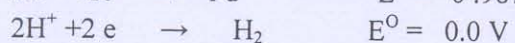
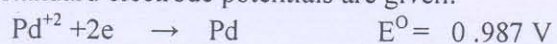
Charotar University of Science and Technology
M.Sc (Advance Organic Chemistry) Semester- I Examination April 2011
OC -703: Physical Chemistry-I
Date: April 27,2011 Time : 10.00 am to 10.30 am Max Marks:20

Part A

1. A chemical reaction has rate constant value of $4.3 \times 10^{-3} \text{ lit.mol}^{-1}.\text{s}^{-1}$. The order of the reaction is : (2)
i.First, ii. Second, iii. Third, iv. Zero
- 2.What is Raoult's law? (2)
3. Ten gram of tin kept at its mp (232°C) is melted. Calculate the entropy change in fusion . Given : latent heat of fusion of tin $=14 \text{ cal.g}^{-1}$. (2)
4. How is the relative population of molecules at different energy levels obtained by Maxwell –Boltzmann distribution? (2)
5. The $T_{1/2}$ of a reaction is halved as the concentration of the reactant is doubled. The order of the reaction is: (2)
i.Zero, ii. First, iii. Third, iv. Second.
6. How is the free energy change ΔG connected to ΔH and ΔS for a chemical reaction at constant temperature T? Discuss the role of ΔH and ΔS on the spontaneity of the reaction. (3)

7. Following standard electrode potentials are given:

(3)



Is it possible to liberate H_2 from HCl by the action of Pd metal?

8. Two solutions are isotonic. Then the solutions have:

(2)

- Same molar concentration and same osmotic pressure.
- Same osmotic pressure but different molar concentration.
- Same molar concentration and different osmotic pressure.

9. The activation energy of a chemical reaction may be obtained if rate constants at different temperatures are known. The equation for this is due to :

(2)

- Einstein, ii. Heisenberg, iii. Arrhenius iv. Schrodinger

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Part B

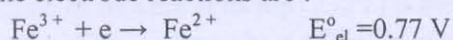
- 1.(a)The consecutive reaction $A \rightarrow B \rightarrow C$ consists of all first order reaction. Derive (7)
 an expression for T_M , the time required to attain maximum concentration of B.
 (b).Show that for a first order reaction, time required for 75 % completion of reaction (3)
 is twice the time for 50% completion.

OR

- (b).For a reaction $A \rightarrow B + C$, it is found that the rate increases by a factor of 2.25, when
 the concentration is increased by a factor of 1.5 at the same temperature. What is the
 order of the reaction?

- 2(a). Set up an electrochemical cell for the reaction: (6)
 $2Fe^{3+}(aq) + Sn^{2+}(aq) \rightarrow 2Fe^{2+}(aq) + Sn^{4+}(aq)$

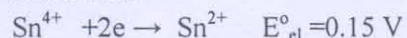
The electrode reactions are :



Calculate the equilibrium constant for the reaction at 25°C.

- (b). Calculate E for the reaction: (4)

$Sn^{4+} + 2e \rightarrow Sn^{2+}$ at 25°C, if the concentration of Sn^{2+} is four times
 that of Sn^{4+} . Given that:



OR

- (b).Calculate the EMF value for the reduction of H^+ at $1.0 \times 10^{-7} \text{ M}$ to H_2 at 1.0 atm.
 Given : $E_{el}^{\circ} = 0.0 \text{ V}$.

- 3(a).Propose a simple scheme for an enzyme catalyzed reaction and derive the (7)
 Michaelis-Menten equation. What is the Physical significance of Michaelis constant
 K_M ?

- (b). A drug has a half life period of 6 h in the body. A patient takes one dose every 6 h. (3)
 What is amount of drug in the patient's body immediately after 6th dose?

OR

- (b).The fast chemical reaction $A \rightleftharpoons B$ is subjected to a sudden temperature. Derive an
 equation that connects relaxation time with the attainment of new equilibrium.

- 4(a). Discuss in brief about the origin of charge on colloidal particles. (3)
 (b). Six moles of an ideal gas expands isothermally and reversibly from a pressure of (4)
 10 atm. to 2 atm. at 300° K. What is the maximum work done? $R=8.314 \text{ J K}^{-1}\text{mol}^{-1}$?

(c). Calculate the change in entropy accompanying the heat of one mole of He gas (3)
assumed ideal from a temperature of 298°K to a temperature of 1000°K at constant
pressure.

5(a). What is Nernst Heat Theorem? What is the important consequence of this theorem? (4)

(b). 100 g of ice at 0°C is dropped into an isolated beaker containing 150 g of water at (4)
 100°C . Calculate Δs for the process that then occurs. Given: Heat capacity of ice is
 $0.5\text{ cal. deg}^{-1}.\text{g}^{-1}$. Heat of fusion of ice = 80 cal. g^{-1} .

OR

(b). Calculate the molar mass of a non volatile solute if at 25°C , the solution containing
 1.6 g. dm^{-3} has an osmotic pressure of 8.3 torr. $R=0.0821\text{ dm}^3.\text{atm. K}^{-1}.\text{mol}^{-1}$.

(c). Write in brief about reverse osmosis.

(2)